



Fundamentals of Optics

Lecturer

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Room 239, South Tower, College of Engineering

Rules

- Final exam: 30%
- Mid-term exam: 30%
- Homework: 25%
- Attendance: 15%

Reference books

Optics, by Eugene Hecht
光学, 赵凯华、钟锡华

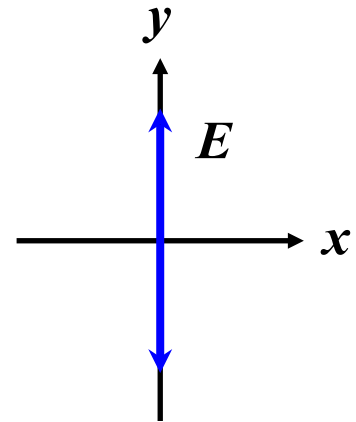
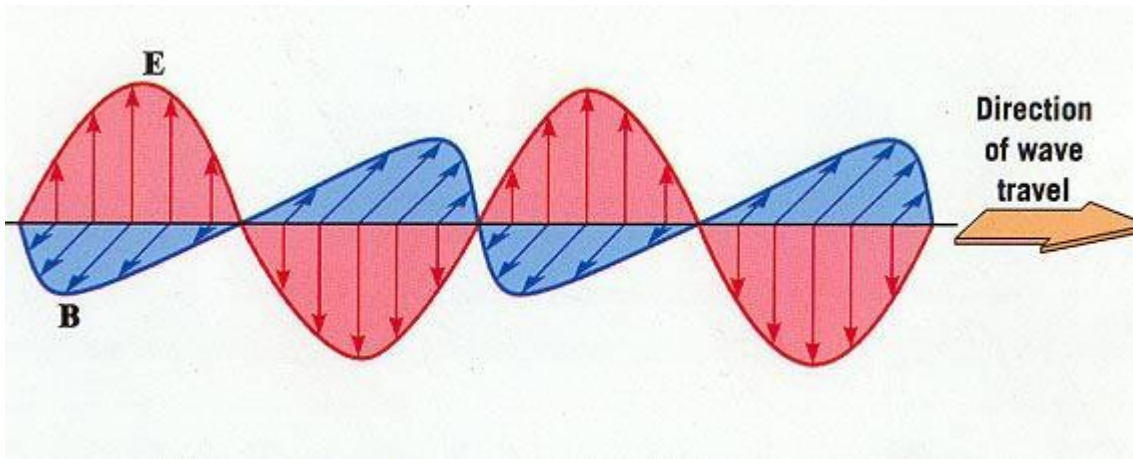
CHAPTER 7

POLARIZATION

7.1 The nature of polarized light

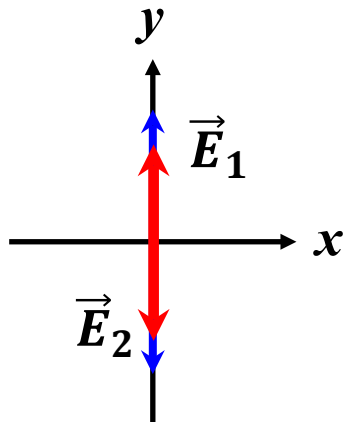
Light: a **transverse electromagnetic wave**.

Linearly polarized or plane-polarized light: the orientation of the electric field is constant, and it resides in **plane-of-vibration**.



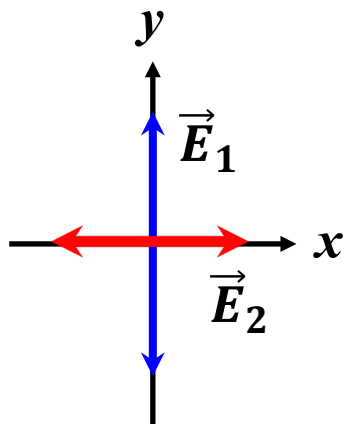
Case: two harmonic, linearly polarized lightwaves of the same frequency, move through the same region of space.

Electrical field vectors are colinear: **superposition**.



Interference, diffraction, beat, ...

Electrical field vectors are perpendicular:



what will happen?

Linear polarization

Two orthogonal optical disturbance:

$$\vec{E}_x(z, t) = \hat{i}E_{0x} \cos(kz - \omega t)$$

$$\vec{E}_y(z, t) = \hat{j}E_{0y} \cos(kz - \omega t + \epsilon)$$

ϵ is the relative phase difference between the waves.

The resultant optical disturbance:

$$\vec{E}(z, t) = \vec{E}_x(z, t) + \vec{E}_y(z, t)$$

Case I: $\epsilon = 2m\pi, m=0, \pm 1, \pm 2, \dots$

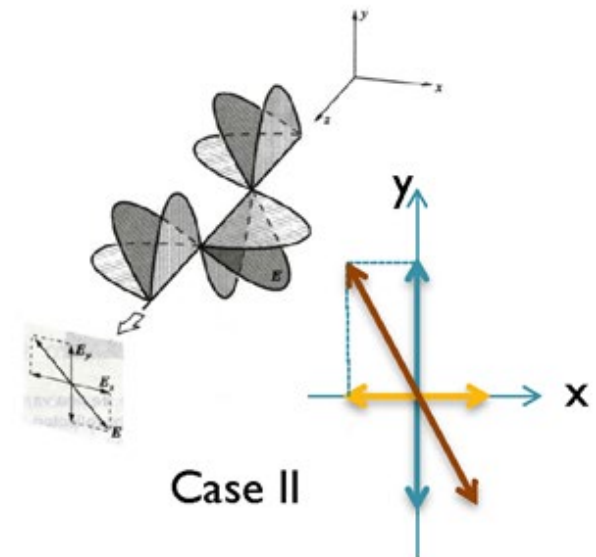
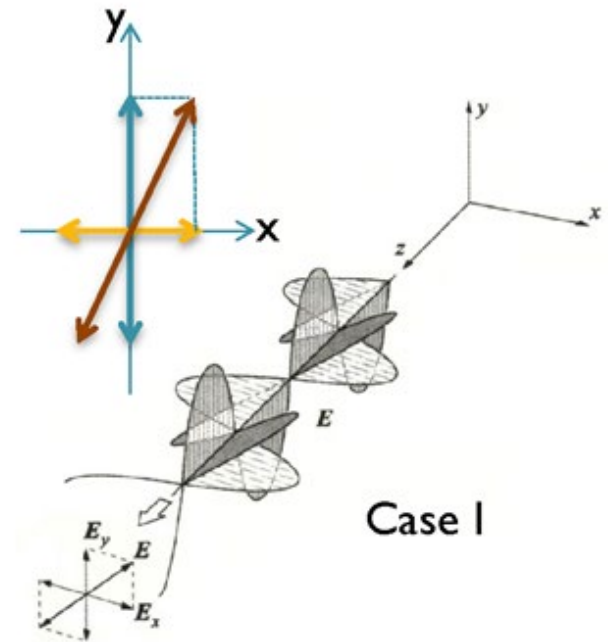
$$\vec{E} = (\hat{i}E_{0x} + \hat{j}E_{0y}) \cos(kz - \omega t)$$

Case II: $\epsilon = (2m+1)\pi, m=0, \pm 1, \pm 2, \dots$

$$\vec{E} = (\hat{i}E_{0x} - \hat{j}E_{0y}) \cos(kz - \omega t)$$

The amplitude is fixed, so it is linearly polarized.

The orientation is tilted.



Circular Polarization

Case III: $\epsilon = -\pi/2 + 2m\pi$, $m=0, \pm 1, \pm 2, \dots$ and $E_{0x} = E_{0y} = E_0$

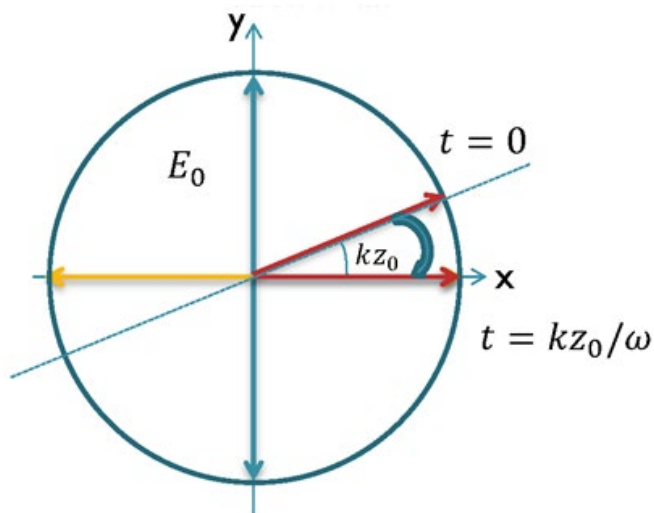
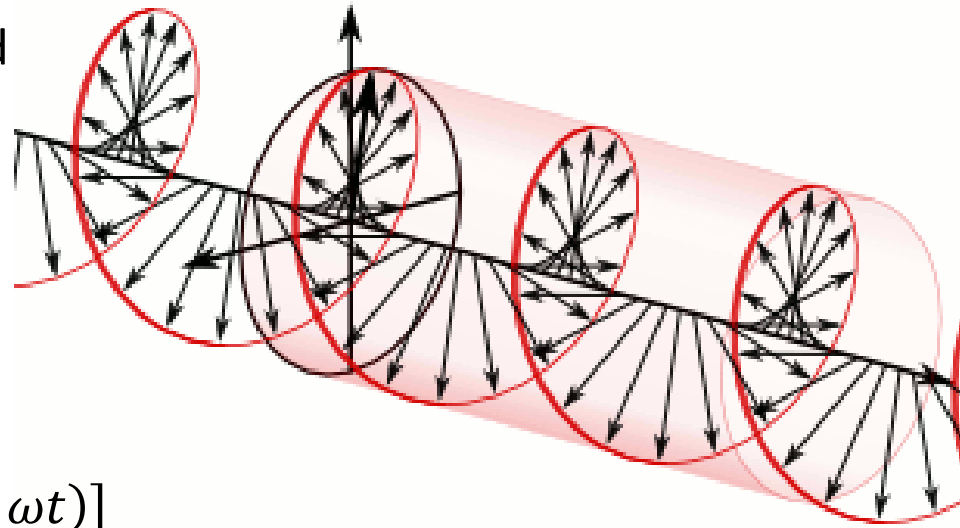
$$\vec{E}_x(z, t) = \hat{i}E_0 \cos(kz - \omega t)$$

$$\vec{E}_y(z, t) = \hat{j}E_0 \sin(kz - \omega t)$$

The consequent wave:

$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)]$$

- ❑ The scalar amplitude is a constant;
- ❑ The direction is time-varying.



For example:

$$t = 0: \vec{E}_x = \hat{i}E_0 \cos kz_0 \text{ and } \vec{E}_y = \hat{j}E_0 \sin kz_0$$

$$t = kz_0/\omega: \vec{E}_x = \hat{i}E_0 \text{ and } \vec{E}_y = 0$$

The resultant electric-field vector $\mathbf{E}$ is rotating clockwise, as seen by an observer toward moving direction of waves, **right-circularly polarized light** (right-circular light)

Case IV: $\epsilon = \pi/2 + 2m\pi$, where $m = 0, \pm 1, \pm 2, \dots$ and $E_{0x} = E_{0y} = E_0$

The consequent wave:

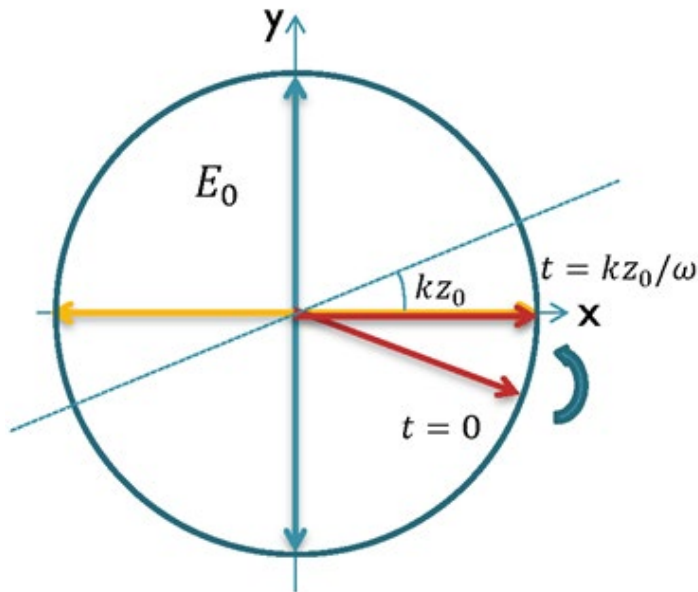
$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)]$$

The resultant electric-field vector **E** rotates counterclockwise,
left-circularly polarized.

For example:

$$t = 0: \vec{E}_x = \hat{i}E_0 \cos kz_0 \text{ and } \vec{E}_y = -\hat{j}E_0 \sin kz_0$$

$$t = kz_0/\omega: \vec{E}_x = \hat{i}E_0 \text{ and } \vec{E}_y = 0$$



❖ A linearly polarized wave can be synthesized from two oppositely polarized circular waves of equal amplitude.

$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)]$$

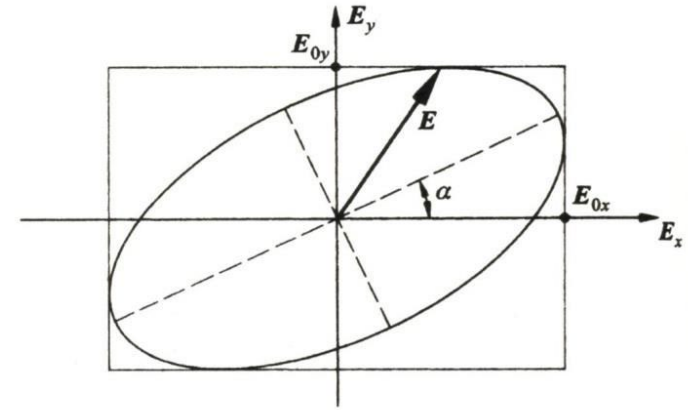
$$\vec{E} = E_0[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)]$$

$$\vec{E} = 2E_0\hat{i} \cos(kz - \omega t)$$

Elliptical polarization

Case V: $E_{0x} \neq E_{0y}$ and ϵ is random

The resultant electric-field vector $\mathbf{E}$ will rotate, and change its magnitude.



$$E_x = E_{0x} \cos(kz - \omega t)$$

$$E_y = E_{0y} \cos(kz - \omega t + \epsilon)$$

Elliptically polarized light

$$\left(\frac{E_y}{E_{0y}}\right)^2 + \left(\frac{E_x}{E_{0x}}\right)^2 - 2\left(\frac{E_x}{E_{0x}}\right)\left(\frac{E_y}{E_{0y}}\right)\cos\epsilon = \sin^2\epsilon$$

This is the equation of an ellipse making an angle α with the (E_x, E_y) -coordinate system

$$\tan 2\alpha = \frac{2E_{0x}E_{0y}\cos\epsilon}{E_{0x}^2 - E_{0y}^2}$$

$$\alpha = 0 \text{ or } \epsilon = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \dots \quad \left(\frac{E_y}{E_{0y}}\right)^2 + \left(\frac{E_x}{E_{0x}}\right)^2 = 1$$

$$E_{0x} = E_{0y} = E_0$$

$$E_x^2 + E_y^2 = E_0^2$$

$$\epsilon = 2m\pi, m = 0, 1, 2, \dots$$

$$E_y = \frac{E_{0y}}{E_{0x}} E_x$$

$$\epsilon = (2m + 1)\pi, m = 0, 1, 2, \dots$$

$$E_y = -\frac{E_{0y}}{E_{0x}} E_x$$

**Elliptically
polarized light**

**Circularly
polarized light**

**Linearly
polarized light**

State of Polarization

Linearly polarized light: $\mathcal{P}$ -state

Right-or left- circular light: $\mathcal{R}$ or $\mathcal{L}$ -state

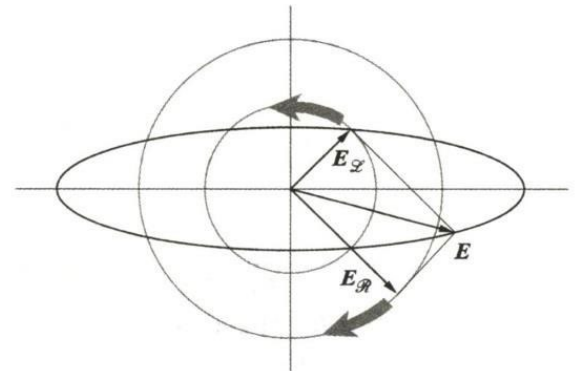
Elliptical polarization: $\mathcal{E}$ -state

$\mathcal{P}$ -state can be represented as a superposition of $\mathcal{R}$ and $\mathcal{L}$ -state.

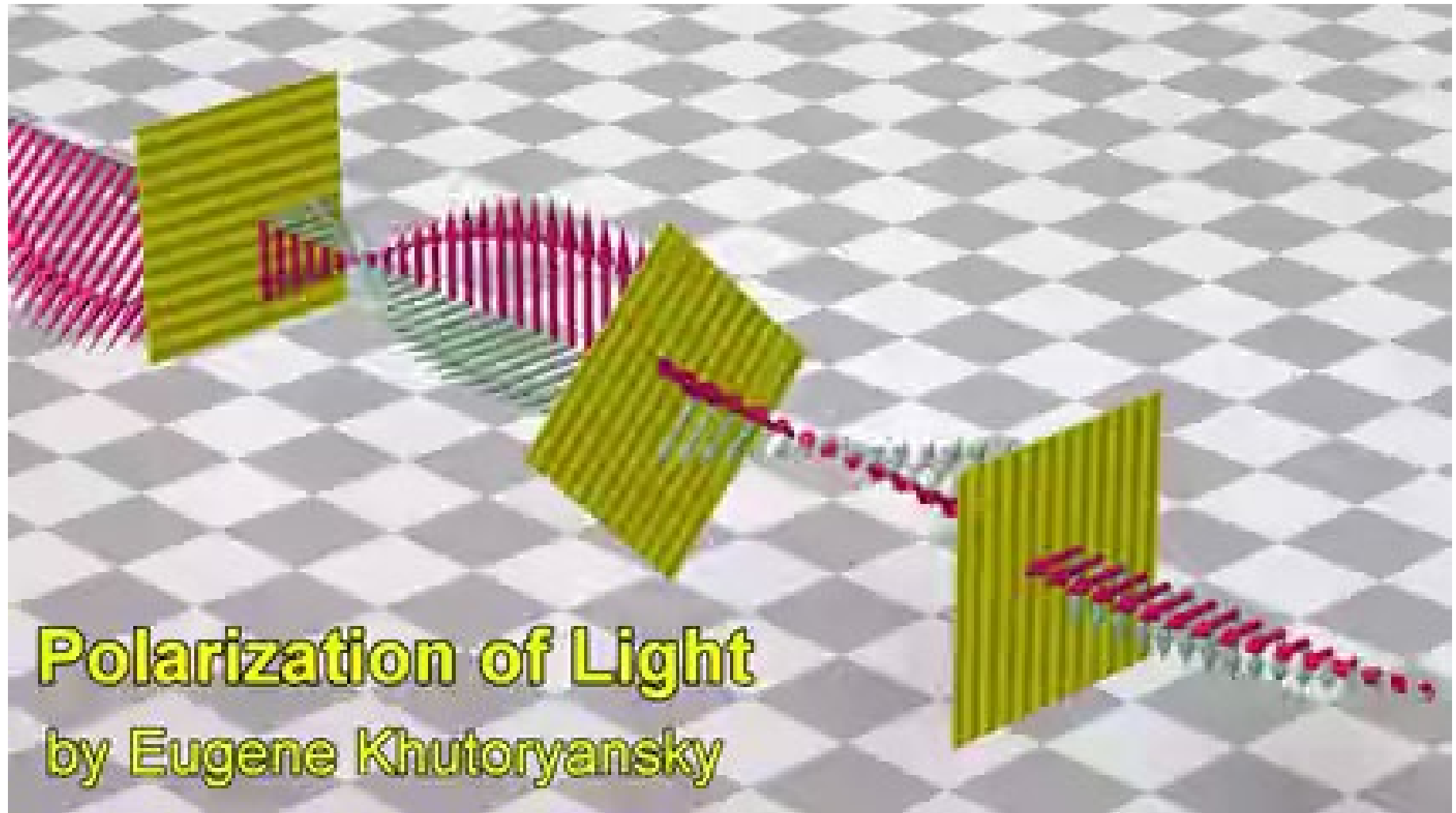
$$\begin{aligned}\vec{E} &= E_0[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)] \\ &\quad + E_0[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)] = 2E_0\hat{i} \cos(kz - \omega t)\end{aligned}$$

$\mathcal{E}$ -state also can be represented as superposition of $\mathcal{R}$ and $\mathcal{L}$ -state with different amplitudes

$$\begin{aligned}\vec{E} &= E_{01}[\hat{i} \cos(kz - \omega t) + \hat{j} \sin(kz - \omega t)] \\ &\quad + E_{02}[\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t)] \\ &= \hat{i}(E_{01} + E_{02}) \cos(kz - \omega t) \\ &\quad + \hat{j}(E_{01} - E_{02}) \sin(kz - \omega t)\end{aligned}$$

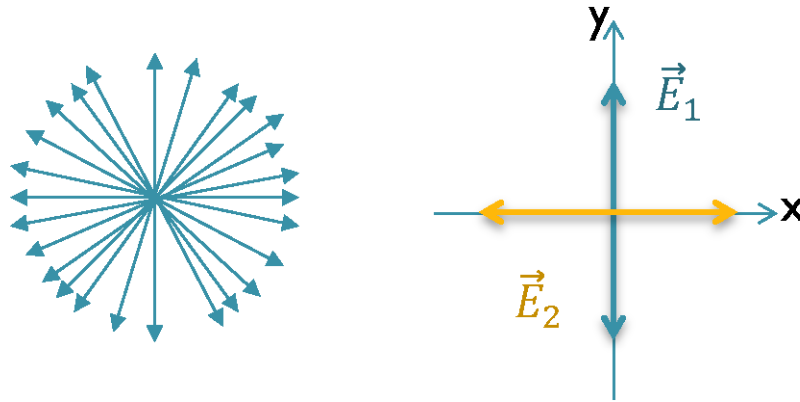


A video introduction about the polarization of light

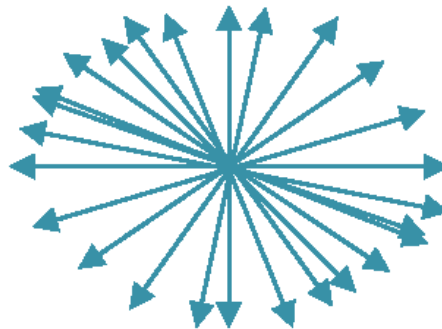


Natural Light

Natural light: the electromagnetic waves emitted from any common source of light are polarized randomly. That is, the electric field changes direction randomly.



Partially polarized waves: light is generally neither completely polarized nor completely unpolarized. (superposition of specific amounts of natural and polarized light).



Angular momentum and the photon picture

Classical picture: electromagnetic wave impinging on an object can impart both energy and linear momentum to that body.

- If the incident plane wave is circularly polarized, the resulting motion is again circular.
- The E-field drives the electron with an angular frequency ω equal to the frequency of the electromagnetic wave.

The power (the energy transferred per unit time)

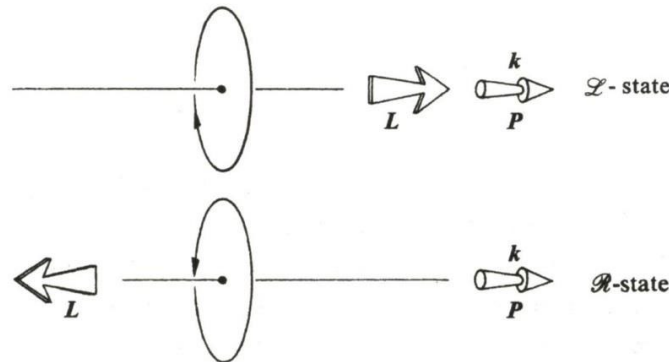
$$\frac{d\xi}{dt} = \omega \mathbf{\Gamma} = \omega \frac{dL}{dt}$$

$$L = \frac{\xi}{\omega}$$

$\mathbf{\Gamma}$: the torque (扭矩);

L : the angular momentum (角动量).

A charge that absorbs a quantity of energy from the incident circular wave will simultaneously absorb an amount of angular momentum。



Quantum-mechanical description:

An electromagnetic wave transfers energy in quantized packets (photons):

$$\xi = h\nu \quad \xi = \hbar\omega \quad (\hbar \equiv \frac{h}{2\pi})$$

The intrinsic or spin angular momentum of a photon is either $-\hbar$ (right-handedness) or $+\hbar$ (left-handedness).

*The spin angular momentum of a photon is completely independent of its energy.

When a charged particle emits or absorbs electromagnetic radiation, it will undergo a change of $\pm\hbar$ in its angular momentum.

The linearly polarized light beam will impart no angular momentum to a target.